AI101 MINI-PROJECT | IIT ROPAR

Smartphone Price Prediction with Linear Regression

Team 39 Predict the price in USD from the phone specifications

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Repo: https://github.com/gopalparihar05/AI-101-Mini-Project-G39-

Problem and data

What and why

- Predict the price (USD) of a smartphone from its specifications.
- A model can show what a fair price is for a phone.
- We use linear regression, the first model of the course.

Dataset

- Source: Kaggle, phone prices from online stores (kaggle.com/datasets/jakubkhalponiak/phones-2024).
- Each row is one phone, in one store, with one storage size.
- Target column: price_usd.
- The same phone is in many rows. We split the data by phone model.

1708
rows

38
columns

469
phone models

14
features used

Data leakage: found and removed

4 columns are made from the price: quantile_10, quantile_50, quantile_90, price_range. A model that uses them sees the answer.

Approach

1 Load and clean
data_prep.py

2 Explore
eda.py

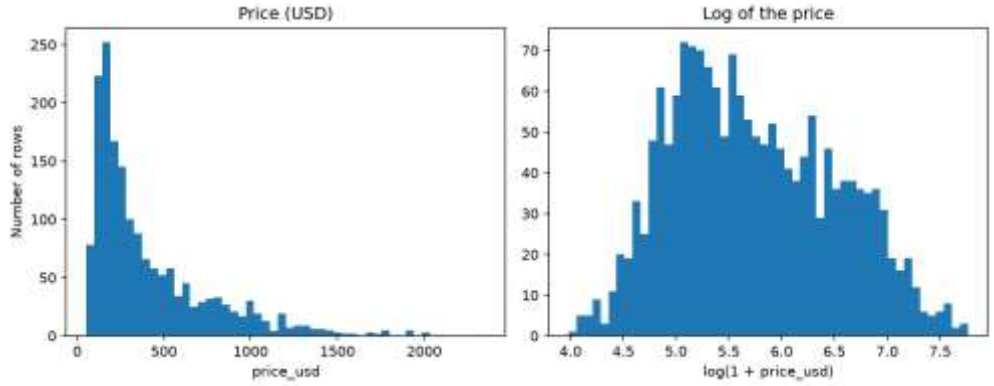
3 Encode and split
train.py

4 Fit 2 models
train.py

5 Evaluate
evaluate.py

Key choices

- Text columns become 0/1 columns (one-hot encoding).
- 80% train, 20% test. All rows of one phone stay on the same side.
- The model learns log(1 + price). The code changes the answer back to USD.
- Simple model: RAM only. Multiple model: 14 features.



Most phones are cheap and a few cost more than 1500 USD. The log price (right) is closer to a bell shape.

Libraries: pandas | numpy | scikit-learn | matplotlib

Results

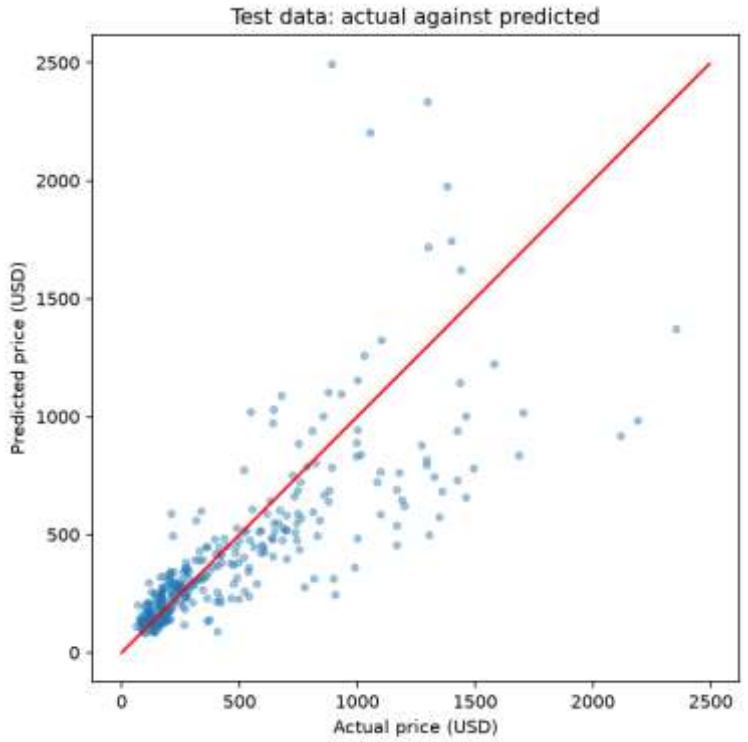
Model (test data)	R2	RMSE (USD)	MAE (USD)
Simple (RAM only)	0.048	404.9	234.7
Multiple (14 features)	0.562	274.6	157.8

Key numbers

- Best model: multiple linear regression, R2 = 0.562.
- A typical error is about 158 USD (MAE). RMSE is 275 USD.
- Test data: 321 rows of phones the model never saw (train: 1387 rows).

```
Step 1/4 (Member 1): load and clean the data
  1708 rows, 16 columns
Step 2/4 (Member 2): exploratory data analysis
  saved eda_summary.txt
  saved 01_price_distribution.png
  saved 02_correlation.png
  saved 03_price_by_brand.png
  saved 04_price_vs_ram.png
Step 3/4 (Member 3): train the linear regression models
  train rows: 1387, test rows: 321
Step 4/4 (Member 4): evaluate on the test data
                                        R2  RMSE_usd  MAE_usd
Simple linear regression (RAM only)      0.048   404.878  234.711
Multiple linear regression (all features)  0.562   274.642  157.813
  saved 05_actual_vs_predicted.png
  saved 06_residuals.png

Done. See the outputs/ folder.
```



Cheap phones are near the red line. Expensive phones are mostly below it: the model predicts them too low.

Learnings

What worked

- We found and removed 4 leakage columns.
- The split by phone model gives an honest test score.
- More features raised R2 from 0.048 to 0.562.
- Storage and RAM go most with the price (correlation 0.68 and 0.62).

What did not work

- RAM alone is weak: R2 is only 0.048.
- The model predicts expensive phones too low.
- 44% of the price change is still not explained.
- Next: add more features, or try a non-linear model.

Who did what

- Nikhil: data cleaning, README
- Rishu: exploration charts, slides
- Pranjal: setup, training, run_all.py
- Gopal: evaluation, video

AI-use disclosure

- ChatGPT (OpenAI): help to understand linear regression, data preparation and debugging.
- Claude (Anthropic): first draft of the Python code, the documentation, these slides and the video scripts.
- The team reviewed and tested the code, ran it on a fresh clone, and checked the results.
- No AI made the dataset or the results. The metrics come from running the code.